

Homework 4 - Choosing Data Formats Deliberately

Goal: Compare common biomedical data formats, identify formatting hazards, and justify a format/storage choice based on the scientific workflow rather than habit.

Part 1 - Compare the same data in three formats (25 points)

- Open `patient_visits.csv`, `patient_visits.xlsx`, and `patient_visits.json`.
- Describe how rows/records, variables/fields, missing values, dates, and identifiers are represented in each.
- State one strength and one limitation of each format for reproducible biomedical analysis.

Part 2 - Diagnose formatting problems (25 points)

Inspect `messy_formatting.csv`. Identify at least five issues that could create incorrect analysis or inconsistent interpretation. For each issue, propose a documented correction rule.

Part 3 - Choose a format for each scenario (25 points)

Scenario	Choose a format/storage approach and justify
A small rectangular analysis dataset that must work in R, Python, and SAS	
An API response with one patient, multiple visits, and multiple labs per visit	
A manually maintained screening log used by two research staff	
A hospital system with millions of encounters, medications, diagnoses, and labs	

Part 4 - Format Conversion Check exercise (20 points)

- Import `patient_visits.csv` into a tool of your choice.
- Export it to another format (for example, XLSX or JSON).
- Re-open the exported file and check participant IDs, dates, missing HbA1c, numeric values, and row count.
- Report any changes or state that no changes were detected after checking.

Synthesis question (5 points)

A collaborator sends you a complex Excel workbook and says, "This is our permanent master dataset." What questions would you ask before treating it as the authoritative source?

Submission

- One Markdown or PDF with Parts 1–3, your round-trip findings, and the synthesis response.
- Include the exported file from Part 4.
- Estimated effort: approximately 60–90 minutes.